

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 38 of 40 — every set uses different data) **Decode the Chart**

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Roll No: 38

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

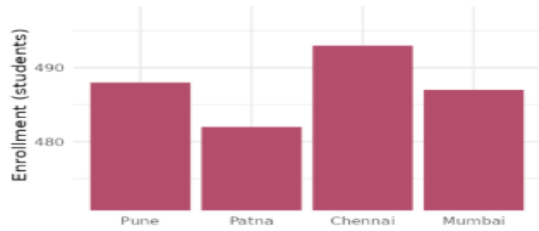


Chart 1



Chart 2

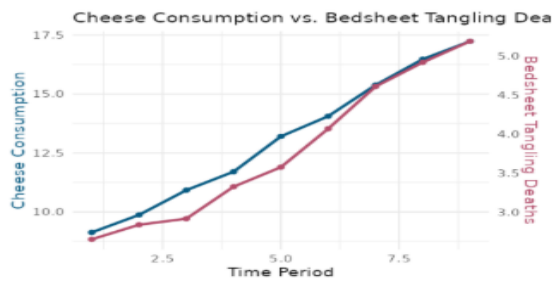


Chart 3

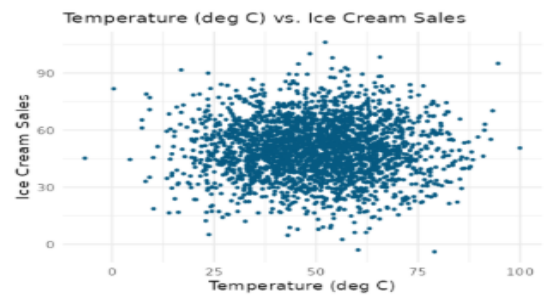


Chart 4

Your notes:

Chart 1 – Bar Chart

What is it trying to tell you?

The bar chart is comparing the student enrollment in four different cities: Pune, Patna, Chennai, and Mumbai. It is showing which city has a higher or lower number of students. From the chart, Chennai appears to have the highest enrollment, while Patna appears to have the lowest.

Do you trust it?

Not completely.

Why?

The main problem is that the y-axis does not start from zero. It starts around 480 instead. Because of this, even a small difference in enrollment appears to be a very large difference in the height of the bars. This can make us think that one city has many more students than another city, even though the actual difference is quite small.

For example, if two cities have enrollment values that are close to each other, starting the axis at zero would show their bars as almost similar in height. Starting the axis at a high value exaggerates the visual difference.

Simple note:

The chart is trying to compare student enrollment between different cities. However, the y-axis does not start from zero, so small differences appear much larger than they actually are. This can give a misleading impression about the difference in enrollment. Therefore, I would be careful

when trusting the visual comparison. A bar chart with the y-axis starting at zero would provide a more accurate comparison.

Chart 2 – Pie Chart

What is it trying to tell you?

The pie chart shows the breakdown of total expenses into different categories. Each slice represents one expense category, and the size of the slice represents its share of the total expenses. A larger slice means that category has a larger share of the expenses.

Do you trust it?

Not very much.

Why?

The main problem is that the pie chart contains too many categories and slices. When there are many slices, it becomes difficult for us to compare their sizes accurately. Some slices are very similar in size, so it is difficult to determine which category is actually larger.

The many different colors also make the chart harder to read. Our eyes are not very good at comparing small differences between angles and areas of pie slices.

A bar chart would be more suitable because we could easily compare the length of each bar and identify the largest and smallest expense categories.

Simple note:

The chart shows how total expenses are divided among different categories. However, there are too many slices, making it difficult to compare the categories accurately. Similar-sized slices can

be confusing, and the different colors do not make the comparison easier. Therefore, I would not completely trust the visual comparison. A bar chart would make the information clearer and easier to compare.

Chart 3 – Dual-Axis Line Chart

What is it trying to tell you?

The chart shows two different measurements changing over a period of time. The two lines appear to increase together as time passes. Because the lines seem to follow a similar pattern, a viewer might think that the two measurements are related to each other.

Do you trust it?

Not completely.

Why?

The problem is that the chart uses two different y-axes. Each measurement has its own scale. The scales can be adjusted in such a way that two completely different patterns appear to move together.

This can create a false impression of correlation. Just because two lines look similar does not necessarily mean that one measurement affects the other or that there is a real relationship between them.

The chart should therefore be interpreted carefully. More evidence or a proper correlation analysis would be needed before saying that the two variables are actually related.

Simple note:

The chart shows two different measurements changing over time, and the lines appear to move together. However, the two measurements use different y-axis scales. These different scales can make unrelated data appear to have a relationship. This may create a false impression of correlation. Therefore, I would not trust the apparent relationship without checking the actual data and using further statistical evidence.

Chart 4 – Scatter Plot

What is it trying to tell you?

The scatter plot shows the relationship between temperature and ice cream sales. Each dot represents one observation. The chart is trying to help us understand whether changes in temperature are associated with changes in ice cream sales.

Do you trust it?

With caution.

Why?

The main problem is that there are too many data points in the chart. Many points are placed very close to each other or directly on top of one another. This is known as overplotting.

Because of overplotting, it becomes difficult to see the actual distribution of the data. We cannot easily tell where the points are most concentrated or whether there is a clear relationship between temperature and ice cream sales.

The chart could be improved by using transparent points, so overlapping points become visible. Other options include using a 2D histogram, hexbin plot, or contour lines.

Simple note:

The chart is trying to show the relationship between temperature and ice cream sales. However, there are a very large number of points, and many of them overlap. This is called overplotting. Because of this, it is difficult to clearly identify the actual pattern or concentration of the data. Therefore, I would interpret the chart with caution. Using transparent points or another suitable visualization would make the data easier to understand.